

MARWADI UNIVERSITY

DEPARTMENT OF AI, ML

DAA (01AI0506) - Lab Practical

Practical 02: Implementation and Time Analysis of Linear and Binary Search

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Aim

Implementation and Time Analysis of Linear and Binary Search as specified in the DAA Lab Manual.

Program

```
#include <iostream>
using namespace std;

int linearSearch(int a[], int n, int key) {
    for (int i = 0; i < n; i++)
        if (a[i] == key) return i;
    return -1;
}

int binarySearch(int a[], int n, int key) {
    int low = 0, high = n - 1;
    while (low <= high) {
        int mid = low + (high - low) / 2;
        if (a[mid] == key) return mid;
        if (a[mid] < key) low = mid + 1;
        else high = mid - 1;
    }
    return -1;
}

int main() {
    int a[] = {10, 25, 7, 40, 15, 30};
    int n = sizeof(a) / sizeof(a[0]);
    cout << "Linear Search (40): " << linearSearch(a, n, 40) << endl;

    int b[] = {5, 10, 15, 20, 25, 30, 35, 40};
    int m = sizeof(b) / sizeof(b[0]);
    cout << "Binary Search (30): " << binarySearch(b, m, 30) << endl;
}
```

Output

```
Linear Search (40): 3
Binary Search (30): 5
```

Time and Space Analysis

Linear Search: Best $\Theta(1)$, Average $\Theta(n)$, Worst $\Theta(n)$, Space $\Theta(1)$. Binary Search: Best $\Theta(1)$, Average $\Theta(\log n)$, Worst $\Theta(\log n)$, iterative Space $\Theta(1)$.

Result

The practical was implemented successfully and the expected output was obtained.